T3 E_+2: full general UV Lagrangian and RGEs

Matchete/RGBeta symbolic output in mathematical notation

E_+2

Additional quartics: 0, additional bilinears: 0, extra Yukawa: none.
General matching and Wilson coefficients are not computed in this symbolic report.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Complete BSM UV Lagrangian

Matchete BSM Lagrangian, excluding the pure Standard Model contribution. $\mathcal{L}_{\text{BSM}} = \text{Coupling(MS1, \{\}, 0)}^2\text{Field(NewScalar1, Scalar, \{Index(d\$}$

$\text{+Field(NewScalar1, Scalar, \{Index(d\$\$1, T3BSMd3)\}, \{Index(d\$\$1, Lorentz)\})Bar(Field(NewScalar1, Scalar, \{Index(d\$\$1, T3BSMd3)\}, \{Index(d\$\$1, Lorentz)\}))}$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal1CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal1CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH1Inv2}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal1CGInv2}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH1Inv2}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal1CGInv2}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{lambdaT3}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{CG}(\text{T3MixCG}, \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Scalar1SelfCGInv1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS1Inv2}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Scalar1SelfCGInv2}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{MS2}, \{\}, 0)^2\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})))$$
$$+\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{Bar}(\text{T3MixCG}), \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$2}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal2CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal2CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv2}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal2CGInv2}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv2}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal2CGInv2}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{lambda12Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3CrossScalarCGInv1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{lambda12Inv2}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3CrossScalarCGInv2}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{lambda12Inv3}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3CrossScalarCGInv3}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS2Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Scalar2SelfCGInv1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS2Inv2}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Scalar2SelfCGInv2}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$-\text{Coupling}(\text{MF}, \{\}, 0)\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

$$+\text{Coupling}(\text{y2}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Y2CG}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(I, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{Proj}(1)), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

$$+\text{Coupling}(\text{y1}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{CG}(\text{T3Y1CG}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(I, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(1)), \text{Transp}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))))$$

$$+i\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{GammaM}(\text{Index}(\text{d\$\$1}, \text{Lorentz}))), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

$$+\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Coupling}(\text{y2}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{CG}(\text{Bar}(\text{T3Y2CG}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{Proj}(-1)), \text{Field}(I, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\}))$$

$$+\text{Bar}(\text{Coupling}(\text{y1}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{Bar}(\text{T3Y1CG}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{NCM}(\text{Transp}(\text{Field}(I, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(-1)), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

Interaction inventory

- `Yukawa1_GenericCG`
- `Yukawa2_GenericCG`
- `Scalar1SelfInv1`
- `Scalar1SelfInv2`
- `Scalar2SelfInv1`
- `Scalar2SelfInv2`
- `HiggsPortal1Inv1`
- `HiggsPortal1Inv2`
- `HiggsPortal2Inv1`
- `HiggsPortal2Inv2`
- `ScalarCrossPortalInv1`
- `ScalarCrossPortalInv2`
- `ScalarCrossPortalInv3`
- `T3ScalarMix_GenericCG`

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{179{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = \frac{{g_2}^3}{6}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = & \, 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = & \, \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} &= 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ &+ 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \frac{3}{4}(\Big(y_1\ (y_1)^\dagger\ Y_e\Big)_{ij} + \Big(y_2\ (y_2)^\dagger\ Y_e\Big)_{ij} + 2\,\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij}) \\ &- \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ &- \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

y_1

$$\begin{aligned}16\pi^2 \frac{dy_1}{d\ln\mu} &= T_\nu^{(1)}\,(y_1)_{ij} \\ &+ \frac{1}{4}(3\,\Big(y_2\ (y_2)^\dagger\ y_1\Big)_{ij} + 2\,\Big(Y_e\ (Y_e)^\dagger\ y_1\Big)_{ij} + 6\,\Big(y_1\ (y_1)^\dagger\ y_1\Big)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(y_1)_{ij} \\ &- \frac{15}{2}{g_1}^2\,(y_1)_{ij}\end{aligned}$$

y_2

$$\begin{aligned}16\pi^2 \frac{dy_2}{d\ln\mu} &= T_\nu^{(2)}\,(y_2)_{ij} \\ &+ \frac{1}{4}(3\,\Big(y_1\ (y_1)^\dagger\ y_2\Big)_{ij} + 2\,\Big(Y_e\ (Y_e)^\dagger\ y_2\Big)_{ij} + 6\,\Big(y_2\ (y_2)^\dagger\ y_2\Big)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(y_2)_{ij} \\ &- \frac{15}{2}{g_1}^2\,(y_2)_{ij}\end{aligned}$$

M_F

$$\begin{aligned}16\pi^2 \frac{dM_F}{d\ln\mu} &= \frac{3}{4}(\Big((y_1)^T\ (y_1)^*\ M_F\Big)_{ij} + \Big(M_F\ (y_2)^\dagger\ y_2\Big)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(M_F)_{ij} \\ &- \frac{27}{2}{g_1}^2\,(M_F)_{ij}\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} &= m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)}) \\ &- 4\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big) \\ &- 12{g_2}^2 m_1^2\end{aligned}$$

$$
\begin{aligned}
&- 6{g_1}^2 m_1^2 \\
&+ m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})
\end{aligned}
$$

m_2^2

$$
\begin{aligned}
16\pi^2 \frac{dm_2^2}{d\ln\mu} = {} & m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)}) \\
& - 4\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big) \\
& - 12{g_2}^2 m_2^2 \\
& - 24{g_1}^2 m_2^2 \\
& + m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})
\end{aligned}
$$

λ_1

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_1}{d\ln\mu} = {} & \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& - 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\
& - 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& - 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\
& + 4\lambda_5\,(\lambda_5)^* \\
& + \frac{9{g_2}^4}{4} \\
& + {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right) \\
& - 9{g_2}^2\lambda_1 \\
& + \frac{3{g_1}^4}{4} \\
& + 12{\lambda_1}^2 \\
& + 6{\lambda_{HS_1}^{(1)}}^2 \\
& + 3{\lambda_{HS_1}^{(A)}}^2 \\
& + 6{\lambda_{HS_2}^{(1)}}^2 \\
& + 3{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = {} & \lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)}) \\
& - \frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big) \\
& + 22{g_2}^4 \\
& + {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_1}^{(1)}\right) \\
& - 24{g_2}^2\lambda_{S_1}^{(1)} \\
& + 12{g_1}^4
\end{aligned}
$$

$$
\begin{aligned}
&+ 6{\lambda_{12}^{(1)}}^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20{\lambda_{12}^{(\times)}}^2}{3} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 14{\lambda_{S_1}^{(1)}}^2 \\
&+ 16{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = {} & \lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 8\lambda_{S_2}^{(A)}) \\
& - \frac{8}{3}\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& + 22{g_2}^4 \\
& + {g_1}^2\left(32{g_2}^2 - 48\lambda_{S_2}^{(1)}\right) \\
& - 24{g_2}^2\lambda_{S_2}^{(1)} \\
& + 192{g_1}^4 \\
& + 6{\lambda_{12}^{(1)}}^2 \\
& + 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
& + \frac{20{\lambda_{12}^{(\times)}}^2}{3} \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + 14{\lambda_{S_2}^{(1)}}^2 \\
& + 16{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 6\operatorname{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\operatorname{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\operatorname{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_1}^{(A)}) \\
& - 2\operatorname{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + 4\lambda_5\,(\lambda_5)^* \\
& + 6{g_2}^4 \\
& - \frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
& + 3{g_1}^4 \\
& - \frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
& + \lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
& + 6\lambda_1\lambda_{HS_1}^{(1)} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$$+ 6{\lambda_{HS_1}^{(A)}}^2$$

$\lambda_{HS_2}^{(1)}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_2}^{(A)}) \\
& - 2\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + 4\lambda_5\,(\lambda_5)^* \\
& + 6{g_2}^4 \\
& - \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\
& + 12{g_1}^4 \\
& - \frac{51{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\
& + \lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
& + 6\lambda_1\lambda_{HS_2}^{(1)} \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
& + 6{\lambda_{HS_2}^{(A)}}^2
\end{aligned}$$

$\lambda_{12}^{(1)}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = {} & \lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}) \\
& - \frac{22}{15}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& + 2\lambda_5\,(\lambda_5)^* \\
& + \frac{63{g_2}^4}{5} \\
& - 24{g_2}^2\lambda_{12}^{(1)} \\
& + 48{g_1}^4 \\
& - 30{g_1}^2\lambda_{12}^{(1)} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + \lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
& + \lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
& + 16{\lambda_{12}^{(A)}}^2 \\
& + 4{\lambda_{12}^{(\times)}}^2 \\
& + 4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
& + 4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} = \lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right) \\
& - \frac{33{g_2}^2\lambda_5}{2} \\
& - \frac{33{g_1}^2\lambda_5}{2} \\
& + 2\lambda_{12}^{(1)}\lambda_5 \\
& + 2\lambda_1\lambda_5 \\
& + 4\lambda_{HS_1}^{(1)}\lambda_5 \\
& + 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = \lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_1}^{(A)}) \\
& - \frac{4}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& - \frac{4}{3}\lambda_5\,(\lambda_5)^* \\
& + {g_1}^2\left(4{g_2}^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right) \\
& - \frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2} \\
& + 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
& + 2\lambda_1\lambda_{HS_1}^{(A)} \\
& + 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
& + 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = \lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_2}^{(A)}) \\
& + \frac{4}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& - \frac{8}{9}\lambda_5\,(\lambda_5)^* \\
& + {g_1}^2\left(8{g_2}^2 - \frac{51\lambda_{HS_2}^{(A)}}{2}\right) \\
& - \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2} \\
& + 12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
& + 2\lambda_1\lambda_{HS_2}^{(A)} \\
& + 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
& + 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{S_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(A)}}{d\ln\mu} = & 4\lambda_{S_1}^{(A)} T_\nu^{(1)} \\
& - \frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big) \\
& - 2{g_2}^4 \\
& + {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_1}^{(A)}\right) \\
& - 24{g_2}^2\lambda_{S_1}^{(A)} \\
& + 12{\lambda_{12}^{(A)}}^2 \\
& - \frac{4{\lambda_{12}^{(\times)}}^2}{3} \\
& + 3{\lambda_{HS_1}^{(A)}}^2 \\
& + 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
& + 20{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} = & 4\lambda_{S_2}^{(A)} T_\nu^{(2)} \\
& - \frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& - 2{g_2}^4 \\
& + {g_1}^2\left(32{g_2}^2 - 48\lambda_{S_2}^{(A)}\right) \\
& - 24{g_2}^2\lambda_{S_2}^{(A)} \\
& + 12{\lambda_{12}^{(A)}}^2 \\
& - \frac{4{\lambda_{12}^{(\times)}}^2}{3} \\
& + 3{\lambda_{HS_2}^{(A)}}^2 \\
& + 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
& + 20{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(A)}}{d\ln\mu} = & \lambda_{12}^{(A)}\left(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3}\right) \\
& + \frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& - \frac{2}{9}\lambda_5\,(\lambda_5)^* \\
& + {g_1}^2\left(16{g_2}^2 - 30\lambda_{12}^{(A)}\right)
\end{aligned}
$$

$$
\begin{aligned}
&- 24{g_2}^2\lambda_{12}^{(A)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)} \\
&+ 3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$\lambda_{12}^{(\times)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(\times)}}{d\ln\mu} = {} & \lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)}) \\
& - \frac{4}{5}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& - \lambda_5\,(\lambda_5)^* \\
& + \frac{51{g_2}^4}{10} \\
& - 24{g_2}^2\lambda_{12}^{(\times)} \\
& - 30{g_1}^2\lambda_{12}^{(\times)} \\
& + 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
& + 10{\lambda_{12}^{(\times)}}^2 \\
& + 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)} \\
& - 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
& + 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)} \\
& - 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}
\end{aligned}
$$